Ahmed Magdi

Database Admin & Passionate Software & DevOps Engineer

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PROFILE

SQL Server database administrator whose work has grown from troubleshooting databases into cloud operations, data-model decisions, integration architecture and reporting to management. Experienced across performance tuning, Azure SQL Managed Instance, backups and high availability, monitoring and incident response, and data engineering.

Leads through technical ownership: brings evidence and clear requirements to developers, network teams and stakeholders, and stays with a problem until the fix is confirmed. Uses AI throughout the work, including Claude connected to SQL Server through MCP.

HOW I WORK

Evidence first
Measure and test before changing anything.

Clear requirements
Pin down what the business needs before designing a fix.

Collaboration
Work with developers, network teams and stakeholders to get changes made.

Follow-through
A fix isn't done until someone confirms it works.

SELECTED ACHIEVEMENTS

- **Customer deletion and restoration.** Agreed a retention approach with stakeholders (keep order lines, set customer references to NULL), then delivered the schema and stored-procedure changes with specific restoration checks.
- **Customer-lookup API performance.** Analysed the workload behind a slow API, rewrote the stored procedure and added a supporting index; the new version was measurably faster in side-by-side tests.
- **Managed Instance connectivity.** Traced failing Azure SQL Managed Instance connections to network routing and gave the network team the details for the route change; they applied it and confirmed connectivity was restored.
- **Automation failure alerting.** Built and tested an alert for failed Azure Automation runs, confirmed email delivery, then tuned it to stop repeat notifications.
- **Claude + MCP for SQL Server.** Built a Model Context Protocol (MCP) server that lets Claude diagnose SQL Server replication issues in a live production environment.

PROJECT EXPERIENCE

Performance and schema design

- Reviewed a proposed column change and its clustered-index design for risk before it went ahead.
- Designed date-based table partitioning so growing tables stay manageable.

Azure SQL Managed Instance

- Planned the move of SQL Server workloads to Azure SQL Managed Instance.
- Scripted and automated database restores on Managed Instance.
- Reviewed maintenance routines and agent-job visibility on Managed Instance.
- Investigated private-endpoint connectivity for workloads running on Azure Kubernetes Service.

Backup, recovery and high availability

- Investigated unreliable backup uploads and proposed a proof of concept to fix them.
- Explored designs for notifying the team automatically when backups complete.
- Investigated data and redo behaviour on an Always On secondary replica.
- Designed transactional replication and subscriber retention for data distribution.

Monitoring and incident response

- Designed detailed query monitoring from the application's point of view, at leadership's request.
- Investigated a reported missing-data incident to establish what happened.
- Validated duplicate localization data, reproduced it with repeated tests, searched the code for its source and brought a focused question to the team standup.

Data engineering and architecture

- Reviewed how the data warehouse loads are orchestrated.
- Specified a refactored data model and ETL design for device installation data.
- Configured runtime credentials for SSIS package deployments.
- Compared architecture options for integrating a third-party system.
- Explored designs for controlling access to sensitive data.
- Prepared performance reporting for stakeholders across organizations.

AI IN MY WORK

Diagnosis	Claude connected to SQL Server through MCP to diagnose live issues.
Code and scripts	T-SQL, PowerShell, SSIS and automation, written and reviewed with AI.
Analysis	Query plans, workloads, logs and incidents, read faster with AI.
Docs and reports	Design comparisons, documentation and stakeholder reports, drafted with AI.

SKILLS

Database	SQL Server, Azure SQL Managed Instance, Always On availability groups, transactional replication, query tuning and indexing, table partitioning, backup and restore, database design, MySQL, PostgreSQL
Data and integration	SSIS, data warehousing and ETL, data modeling, integration architecture, data access design
Cloud and DevOps	Azure, Azure Automation, Azure Kubernetes Service, Docker, Kubernetes, Terraform, CI/CD pipelines, Git, PowerShell, Bash
Software engineering	.NET, C#, ASP.NET Core, Blazor, JavaScript, TypeScript, RESTful APIs